

Module 5

Bode Plots

Bode plot is a frequency response plot of the sinusoidal transfer function of a system.

The Bode plot or the Bode diagram consists of two plots –

- Magnitude plot
- Phase plot

In both the plots, x-axis represents angular frequency (logarithmic scale). Whereas, yaxis represents the magnitude (linear scale) of open loop transfer function in the magnitude plot and the phase angle (linear scale) of the open loop transfer function in the phase plot.

The **magnitude** of the open loop transfer function in dB is -

$$M = 20 \log |G(j\omega)H(j\omega)|$$

The **phase angle** of the open loop transfer function in degrees is -

$$\phi = \angle G(j\omega)H(j\omega)$$

Basic of Bode Plots

The following table shows the slope, magnitude and the phase angle values of the terms present in the open loop transfer function. This data is useful while drawing the Bode plots.

Type of term	$G(j\omega)H(j\omega)$	Slope(dB/dec)	Magnitude (dB)	Phase angle(degrees)
Constant	K	0	$20 \log K$	0
Zero at origin	$j\omega$	20	$20 \log \omega$	90
'n' zeros at origin	$(j\omega)^n$	$20 n$	$20 n \log \omega$	$90 n$

Pole at origin	$\frac{1}{j\omega}$	-20	$-20 \log \omega$	-90 or 270
'n' poles at origin	$\frac{1}{(j\omega)^n}$	-20 n	$-20 n \log \omega$	-90 n or 270 n
Simple zero	$1 + j\omega r$	20	$0 \text{ for } \omega < \frac{1}{r}$ $20 \log \omega r \text{ for } \omega > \frac{1}{r}$	$0 \text{ for } \omega < \frac{1}{r}$ $90 \text{ for } \omega > \frac{1}{r}$
Simple pole	$\frac{1}{1 + j\omega r}$	-20	$0 \text{ for } \omega < \frac{1}{r}$ $-20 \log \omega r \text{ for } \omega > \frac{1}{r}$	$0 \text{ for } \omega < \frac{1}{r}$ $-90 \text{ or } 270 \text{ for } \omega > \frac{1}{r}$
Second order derivative term	$\omega_n^2 \left(1 - \frac{\omega^2}{\omega_n^2} + \frac{2j\delta\omega}{\omega_n} \right)$	40	$40 \log \omega_n \text{ for } \omega < \omega_n$ $20 \log (2\delta\omega_n^2) \text{ for } \omega = \omega_n$	$0 \text{ for } \omega < \omega_n$ $90 \text{ for } \omega = \omega_n$
Second order integral term	$\frac{1}{\omega_n^2 \left(1 - \frac{\omega^2}{\omega_n^2} + \frac{2j\delta\omega}{\omega_n} \right)}$	-40	$-40 \log \omega_n \text{ for } \omega < \omega_n$ $-20 \log (2\delta\omega_n^2) \text{ for } \omega = \omega_n$ $-40 \log \omega \text{ for } \omega > \omega_n$	$-0 \text{ for } \omega < \omega_n$ $-90 \text{ for } \omega = \omega_n$ $-180 \text{ for } \omega > \omega_n$

Consider the open loop transfer function $G(s)H(s) = K$.

Magnitude $M = 20 \log K$ dB

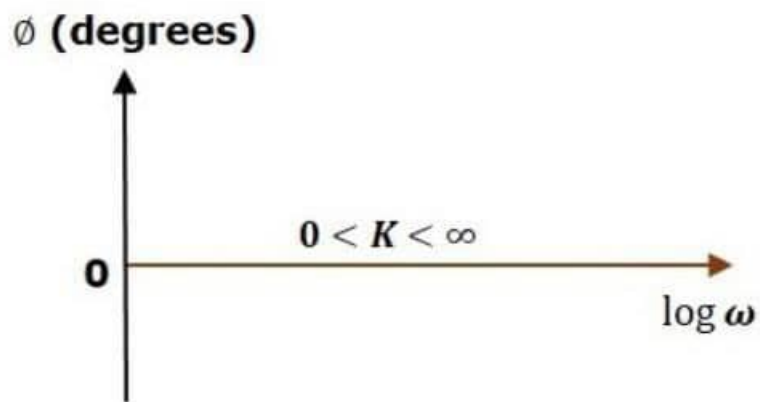
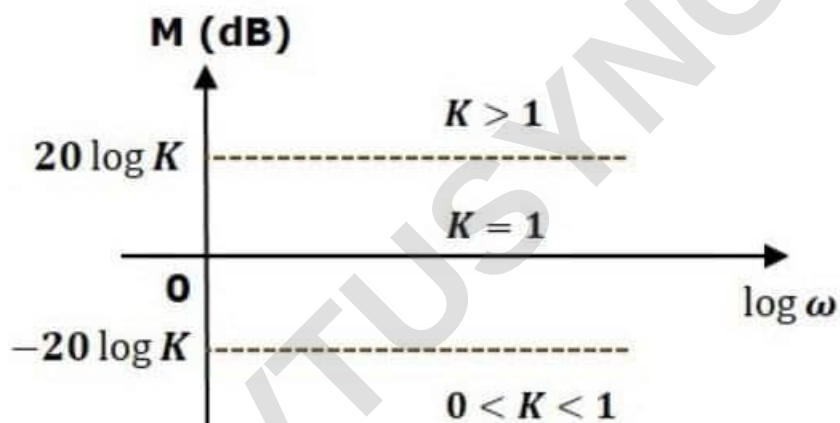
Phase angle $\phi = 0$ degrees

If $K = 1$, then magnitude is 0 dB.

If $K > 1$, then magnitude will be positive.

If $K < 1$, then magnitude will be negative.

The following figure shows the corresponding Bode plot.



The magnitude plot is a horizontal line, which is independent of frequency. The 0 dB line itself is the magnitude plot when the value of K is one. For the positive values of K , the horizontal line will shift $20 \log K$ dB above the 0 dB line. For the negative values of K , the horizontal line will shift $20 \log K$ dB below the 0 dB line. The Zero degrees line itself is the phase plot for all the positive values of K .

Consider the open loop transfer function $G(s)H(s) = s$.

Magnitude $M = 20 \log \omega$ dB

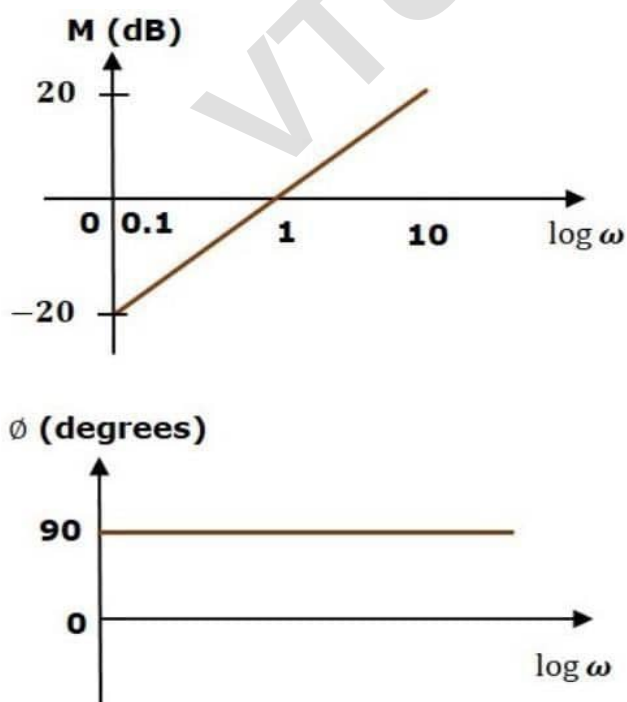
Phase angle $\phi = 90^\circ$

At $\omega = 0.1$ rad/sec, the magnitude is -20 dB.

At $\omega = 1$ rad/sec, the magnitude is 0 dB.

At $\omega = 10$ rad/sec, the magnitude is 20 dB.

The following figure shows the corresponding Bode plot.



The magnitude plot is a line, which is having a slope of 20 dB/dec. This line started at $\omega = 0.1$ rad/sec having a magnitude of -20 dB and it continues on the same slope. It is touching 0 dB line at $\omega = 1$ rad/sec. In this case, the phase plot is 90° line.

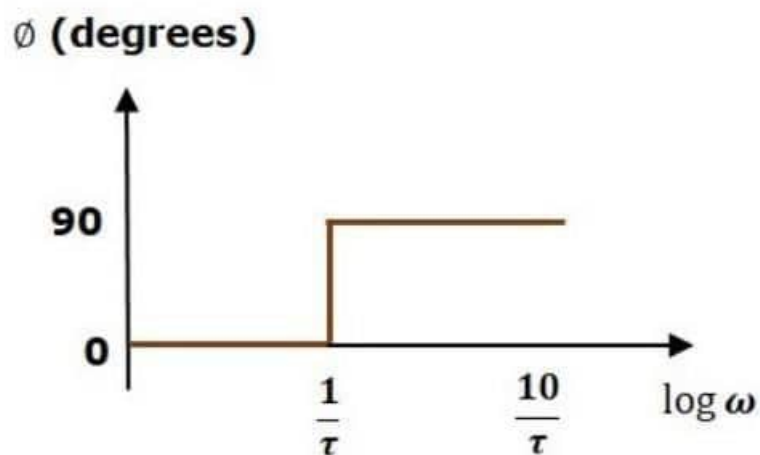
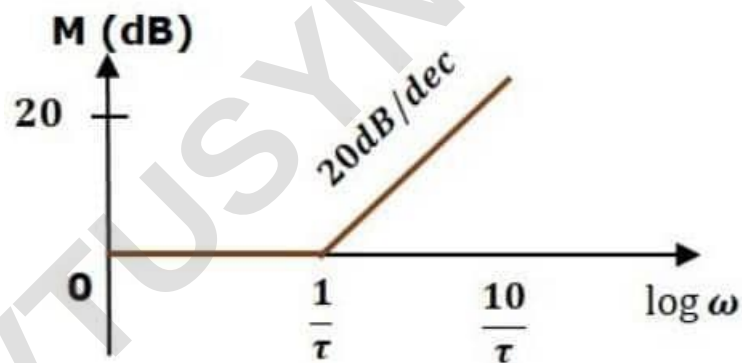
Consider the open loop transfer function $G(s)H(s) = 1 + s\tau$.

Magnitude $M = 20 \log \sqrt{1 + \omega^2 \tau^2}$ dB

Phase angle $\phi = \tan^{-1} \omega \tau$ degrees

For $\omega < \frac{1}{\tau}$, the magnitude is 0 dB and phase angle is 0 degrees.

For $\omega > \frac{1}{\tau}$, the magnitude is $20 \log \omega \tau$ dB and phase angle is 90° .



The magnitude plot is having magnitude of 0 dB upto $\omega = \frac{1}{\tau}$ rad/sec. From $\omega = \frac{1}{\tau}$ rad/sec, it is having a slope of 20 dB/dec. In this case, the phase plot is having phase angle of 0 degrees up to $\omega = \frac{1}{\tau}$ rad/sec and from here, it is having phase angle of 90° . This Bode plot is called the **asymptotic Bode plot**.

Rules for Construction of Bode Plots

Follow these rules while constructing a Bode plot.

- Represent the open loop transfer function in the standard time constant form.
- Substitute, $s=j\omega$ in the above equation.
- Find the corner frequencies and arrange them in ascending order.
- Consider the starting frequency of the Bode plot as $1/10^{\text{th}}$ of the minimum corner frequency or 0.1 rad/sec whichever is smaller value and draw the Bode plot upto 10 times maximum corner frequency.
- Draw the magnitude plots for each term and combine these plots properly.
- Draw the phase plots for each term and combine these plots properly.

Problems

(1) Plot Bode diagram for the following transfer function and obtain the gain and phase cross over Frequencies.

$$\frac{10}{s(1 + 0.4s)(1 + 0.1s)}$$

Solution: The sinusoidal transfer function $G(j\omega)$ is obtained by replacing s by $j\omega$ in the given transfer function

$$G(j\omega) = \frac{10}{j\omega(1 + j0.4\omega)(1 + j0.1\omega)}$$

Magnitude Plot

Corner frequencies are

$$\omega_{c1} = \frac{1}{0.4} = 2.5 \text{ rad/sec and } \omega_{c2} = \frac{1}{0.1} = 10 \text{ rad/sec}$$

Term	Corner frequency rad/sec	Slope db/dec	Change in slope db/dec
$\frac{10}{j\omega}$	—	-20	
$\frac{1}{1+j0.4\omega}$	$\omega_{c1} = \frac{1}{0.4} = 2.5$	-20	-20 - 20 = -40
$\frac{1}{1+j0.1\omega}$	$\omega_{c2} = \frac{1}{0.1} = 10$	-20	-40 - 20 = -60

Choose a low frequency ω_l such that $\omega_l < \omega_{c1}$ and choose a high frequency ω_h such that $\omega_h > \omega_{c2}$.

Let, $\omega_l = 0.1 \text{ rad/sec}$, and $\omega_h = 50 \text{ rad/sec}$.

Let, $A = |G(j\omega)|$ in db.

Let us calculate A at ω_l , ω_{c1} , ω_{c2} and ω_h .

$$\text{At } \omega = \omega_l, A = 20 \log \left| \frac{10}{j\omega} \right| = 20 \log \frac{10}{0.1} = 40 \text{ db}$$

$$\text{At } \omega = \omega_{c1}, A = 20 \log \left| \frac{10}{j\omega} \right| = 20 \log \frac{10}{2.5} = 12 \text{ db}$$

$$\text{At } \omega = \omega_{c2}, A = \left[\text{Slope from } \omega_{c1} \text{ to } \omega_{c2} \times \log \frac{\omega_{c2}}{\omega_{c1}} \right] + A_{(\text{at } \omega = \omega_{c1})} = -40 \times \log \frac{10}{2.5} + 12 = -12 \text{ db}$$

$$\text{At } \omega = \omega_h, A = \left[\text{Slope from } \omega_{c2} \text{ to } \omega_h \times \log \frac{\omega_h}{\omega_{c2}} \right] + A_{(\text{at } \omega = \omega_{c2})} = -60 \times \log \frac{50}{10} + (-12) = -54 \text{ db}$$

PHASE PLOT

The phase angle of $G(j\omega)$ as a function of ω is given by,

$$\phi = -90^\circ - \tan^{-1} 0.4\omega - \tan^{-1} 0.1\omega$$

ω rad/sec	$\tan^{-1} 0.4 \omega$ deg	$\tan^{-1} 0.1 \omega$ deg	$\phi = \angle G(j\omega)$ deg	Points in phase plot
0.1	2.29	0.57	$-92.86 \approx -92$	e
1	21.80	5.71	$-117.5 \approx -118$	f
2.5	45.0	14.0	$-149 \approx -150$	g
4	57.99	21.8	$-169.79 \approx -170$	h
10	75.96	45.0	$-210.96 \approx -210$	i
20	82.87	63.43	$-236.3 \approx -236$	j

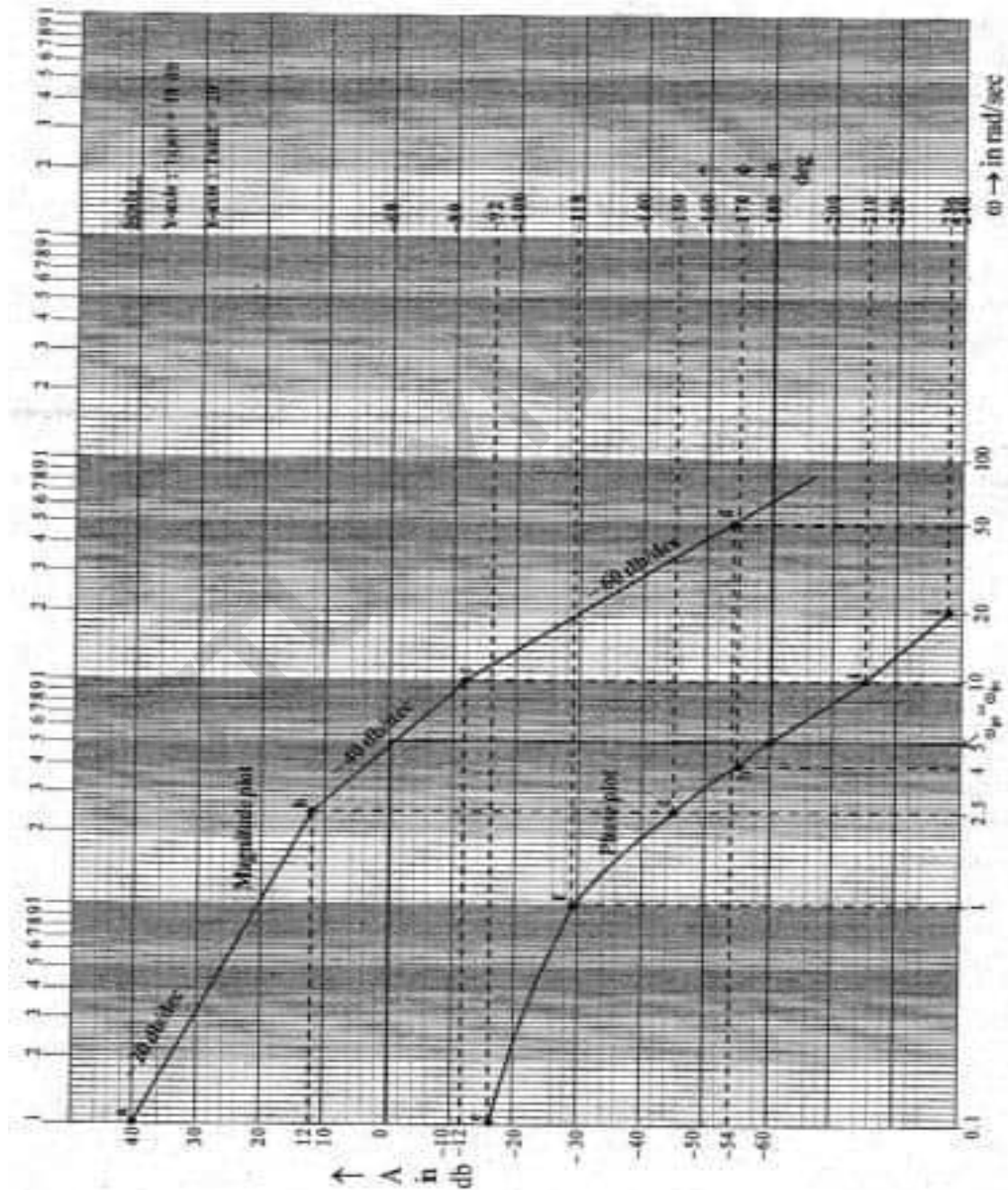


Fig 3.4.1 : Bode plot of transfer function, $G(s) = \frac{10}{s(1 + 0.4s)(1 + 0.1s)}$

Result: Gain cross-over frequency = 5 rad/sec Phase cross-over frequency = 5 rad/sec

(2) Plot Bode diagram for the following transfer function and obtain the cross over frequency.

$$\frac{20}{s(1+3s)(1+4s)}$$

Solution: The sinusoidal transfer function $G(j\omega)$ is obtained by replacing s by $j\omega$ in the given transfer function

$$G(j\omega) = \frac{20}{j\omega(1+j3\omega)(1+j4\omega)}$$

Magnitude Plot

Corner frequencies are

The corner frequencies are, $\omega_{c1} = \frac{1}{4} = 0.25 \text{ rad/sec}$, $\omega_{c2} = \frac{1}{3} = 0.333 \text{ rad/sec}$.

Term	Corner frequency rad/sec	Slope db/dec	Change in slope db/dec
$\frac{20}{j\omega}$	—	-20	
$\frac{1}{1+j4\omega}$	$\omega_{c1} = \frac{1}{4} = 0.25$	-20	-20 - 20 = -40
$\frac{1}{1+j3\omega}$	$\omega_{c2} = \frac{1}{3} = 0.33$	-20	-40 - 20 = -60

Choose a frequency ω_l such that $\omega_l < \omega_{c1}$ and choose a high frequency ω_h such that $\omega_h > \omega_{c2}$.

Let, $\omega_l = 0.15 \text{ rad/sec}$ and $\omega_h = 1 \text{ rad/sec}$.

Let, $A = |G(j\omega)|$ in db.

Let us calculate A at ω_l , ω_{c1} , ω_{c2} and ω_h .

$$\text{At } \omega = \omega_l, A = |G(j\omega)| = 20 \log \left| \frac{20}{0.15} \right| = 42.5 \text{ db}$$

$$\text{At } \omega = \omega_{c1}, A = |G(j\omega)| = 20 \log \left| \frac{20}{0.25} \right| = 38 \text{ db}$$

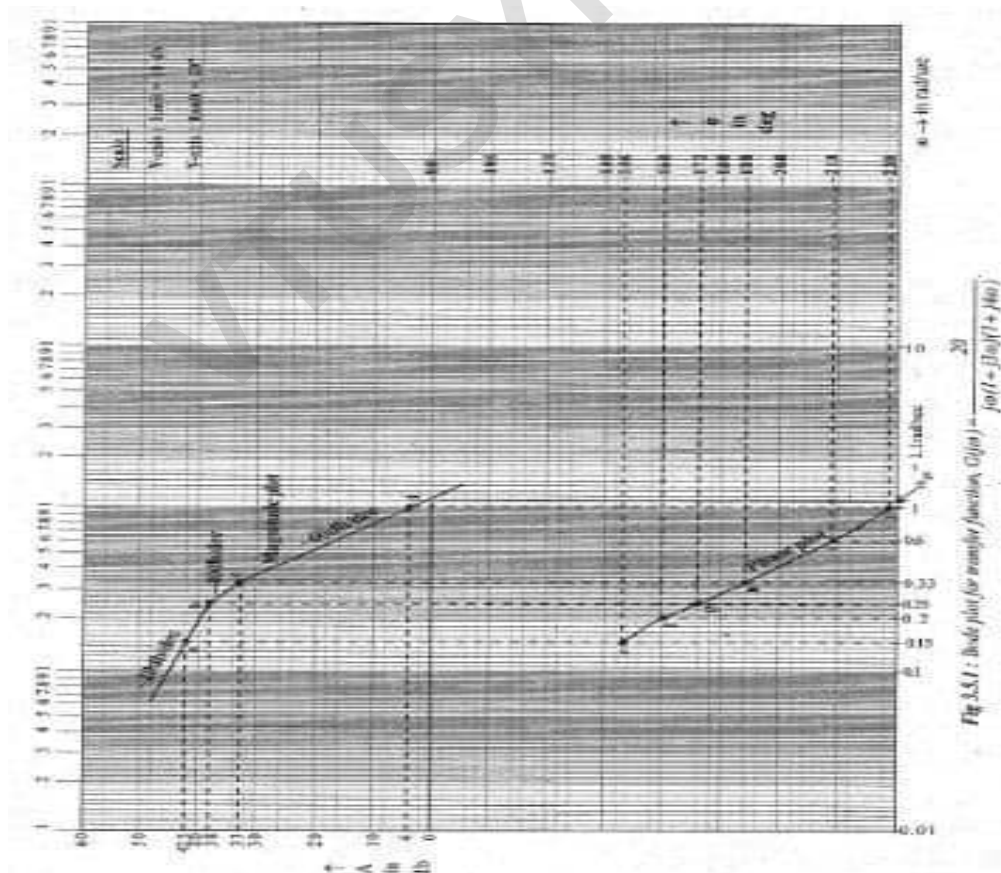
$$\begin{aligned} \text{At } \omega = \omega_{c2}, A &= \left[\text{Slope from } \omega_{c1} \text{ to } \omega_{c2} \times \log \frac{\omega_{c2}}{\omega_{c1}} \right] + A_{(\text{at } \omega = \omega_{c1})} \\ &= -40 \times \log \frac{0.33}{0.25} + 38 = 33 \text{ db} \end{aligned}$$

$$\begin{aligned} \text{At } \omega = \omega_h, A &= \left[\text{Slope from } \omega_{c2} \text{ to } \omega_h \times \log \frac{\omega_h}{\omega_{c2}} \right] + A_{(\text{at } \omega = \omega_{c2})} \\ &= -60 \times \log \frac{1}{0.33} + 33 = 4 \text{ db} \end{aligned}$$

PHASE PLOT

The phase angle of $G(j\omega)$, $\phi = -90^\circ - \tan^{-1} 3\omega - \tan^{-1} 4\omega$

ω , rad/sec	$\tan^{-1} 3\omega$, deg	$\tan^{-1} 4\omega$, deg	$\phi = \angle G(j\omega)$, deg	Points in phase plot
0.15	24.22	30.96	$-145.18 \approx -146$	e
0.2	30.96	38.66	$-159.61 \approx -160$	f
0.25	36.86	45.0	$-171.86 \approx -172$	g
0.33	44.7	52.8	$-187.5 \approx -188$	h
0.6	60.14	67.38	$-218.32 \approx -218$	i
1	71.56	75.96	$-237.56 \approx -238$	j



Result : From the graph the gain cross over frequency is $\omega_{gc} = 1.1 \text{ rad/sec}$

3)

For the function, $G(s) = \frac{5(1+2s)}{(1+4s)(1+0.25s)}$, draw the bode plot.

SOLUTION

The sinusoidal transfer function $G(j\omega)$ is obtained by replacing s by $j\omega$ in $G(s)$.

$$\therefore G(j\omega) = \frac{5(1+j2\omega)}{(1+j4\omega)(1+j0.25\omega)}$$

MAGNITUDE PLOT

The corner frequencies are, $\omega_{c1} = \frac{1}{4} = 0.25 \text{ rad/sec}$, $\omega_{c2} = \frac{1}{2} = 0.5 \text{ rad/sec}$, $\omega_{c3} = \frac{1}{0.25} = 4 \text{ rad/sec}$

The various terms of $G(j\omega)$ are listed in table-1 in the increasing order of their corner frequencies. Also the table shows the slope contributed by the each term and the change in slope at the corner frequency.

Choose a low frequency ω_l such that $\omega_l < \omega_{c1}$ and choose a high frequency ω_h such that $\omega_h > \omega_{c3}$. Let $\omega_l = 0.1 \text{ rad/sec}$ and $\omega_h = 10 \text{ rad/sec}$.

Let $A = |G(j\omega)|$ in db and let us calculate A at $\omega_l, \omega_{c1}, \omega_{c2}, \omega_{c3}$ and ω_h .

Term	Corner frequency rad/sec	Slope db/dec	Change in slope db/deg
5	—	0	—
$\frac{1}{1+j4\omega}$	$\omega_{c1} = \frac{1}{4} = 0.25$	-20	$0 - 20 = -20$
$1+j2\omega$	$\omega_{c2} = \frac{1}{2} = 0.5$	20	$-20 + 20 = 0$
$\frac{1}{1+j0.25\omega}$	$\omega_{c3} = \frac{1}{0.25} = 4$	-20	$0 - 20 = -20$

At $\omega = \omega_l$, $A = |G(j\omega)| = 20 \log 5 = +14 \text{ db}$

At $\omega = \omega_{c1}$, $A = |G(j\omega)| = 20 \log 5 = +14 \text{ db}$

At $\omega = \omega_{c2}$, $A = \left[\text{Slope from } \omega_{c1} \text{ to } \omega_{c2} \times \log \frac{\omega_{c2}}{\omega_{c1}} \right] + A_{(\text{at } \omega = \omega_{c1})} = -20 \times \log \frac{0.5}{0.25} + 14 = +8 \text{ db}$

At $\omega = \omega_{c3}$, $A = \left[\text{Slope from } \omega_{c2} \text{ to } \omega_{c3} \times \log \frac{\omega_{c3}}{\omega_{c2}} \right] + A_{(\text{at } \omega = \omega_{c2})} = 0 \times \log \frac{4}{0.5} + 8 = +8 \text{ db}$

$$\text{At } \omega = \omega_n, A = \left[\text{Slope from } \omega_{c3} \text{ to } \omega_n \times \log \frac{\omega_n}{\omega_{c3}} \right] + A_{(\text{at } \omega = \omega_{c3})} = -20 \log \frac{10}{4} + 8 = 0 \text{ db}$$

PHASE PLOT

The phase angle of $G(j\omega)$, $\phi = \tan^{-1}(2\omega) - \tan^{-1}(4\omega) - \tan^{-1}(0.25\omega)$

ω	$\tan^{-1} 2\omega$ deg	$\tan^{-1} 4\omega$ deg	$\tan^{-1} 0.25\omega$ deg	$\phi = \angle G(j\omega)$	Points in phase plot
0.1	11.3	21.8	1.43	$-11.93 \approx -12$	f
0.25	26.56	45.0	3.5	$-21.94 \approx -22$	g
0.5	45.0	63.43	7.1	$-25.53 \approx -26$	h
2	75.96	82.87	26.56	$-33.47 \approx -33$	i
4	82.87	86.42	45.0	$-48.55 \approx -49$	j
10	87.13	88.56	68.19	$-69.62 \approx -70$	k
50	89.42	89.71	85.42	$-85.71 \approx -86$	l

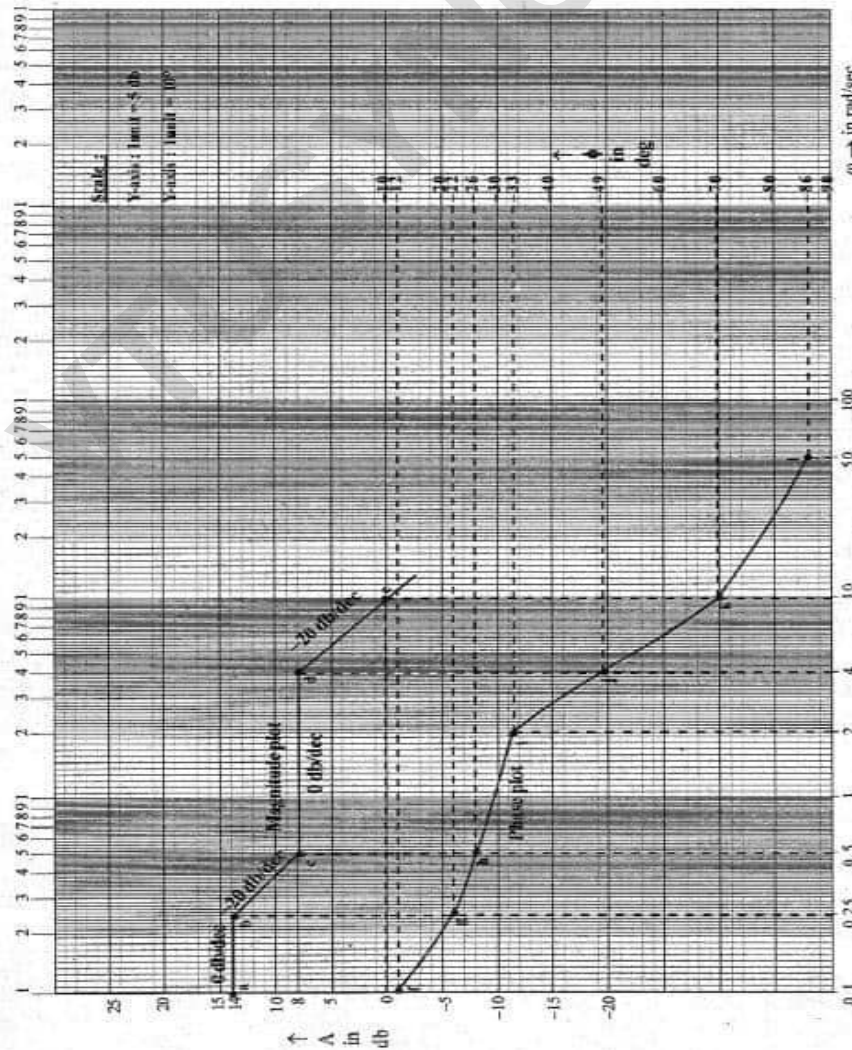
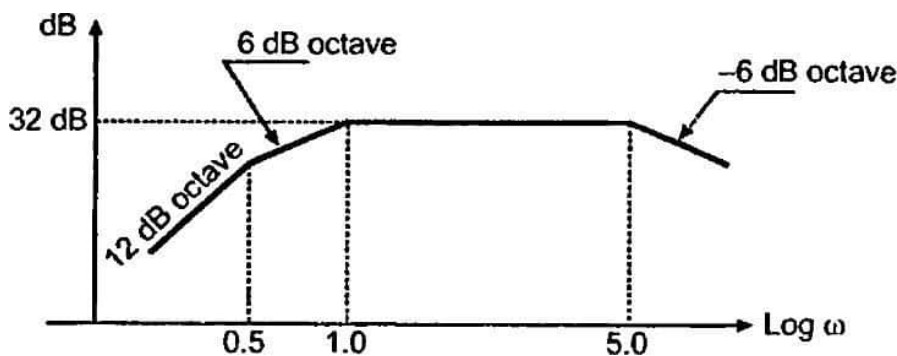


Fig 3.6.1 : Bode plot of transfer function, $G(j\omega) = \frac{5(1+j2\omega)}{(1+j4\omega)(1+j0.25\omega)}$

For the plot shown below determine the transfer function.



Solution : The slope dB/octave is to be converted to dB/decade. Initial slope is +12 dB/octave, hence there are two zeros at the origin. This is because

$$6 \text{ dB/octave} = 20 \text{ dB/decade}$$

$$12 \text{ dB/octave} = 40 \text{ dB/decade}$$

Now the equation of line after $\omega = 0.5$ is say

$$M = 20 \log \omega + C$$

at $\omega = 1, M = +32 \text{ dB}$ shown

$$\therefore 32 = 20 \log 1 + C$$

$$\therefore C = 32$$

At $\omega = 0.5, M = 20 \log 0.5 + 32 = +26 \text{ dB}$

Now $\omega = 0.5, M = 26 \text{ dB}$ is also on the initial line whose equation is

$$M = +40 \log \omega + C_1$$

At $\omega = 0.5, +26 = 40 \log 0.5 + C_1$

$$\therefore C_1 = 38.0412 \text{ dB}$$

Now this line must have $M = 0 \text{ dB}$ at $\omega = 1$ for $K = 1$. But at $\omega = 1$, $M = 40 \log 1 + 38.0412$

i.e. $M = 38.0412 \text{ dB}$

This is due to contribution of system gain constant K.

$$\therefore 20 \log K = 38.0412$$

$$\therefore K = 79.8$$

At $\omega_C = 0.5$, slope changed by -20 , there is simple pole.

$$\begin{aligned} \text{Factor} &= \frac{1}{(1 + T_1 s)}, \text{ where } T_1 = \frac{1}{\omega_C} = \frac{1}{0.5} = 2 \\ &= \frac{1}{(1 + 2s)} \end{aligned}$$

At $\omega_C = 1$, slope changed by -20 , there is simple pole.

$$\begin{aligned} \text{Factor} &= \frac{1}{1 + T_2 s}, \text{ where } T_2 = \frac{1}{\omega_C} = \frac{1}{1} = 1 \\ &= \frac{1}{1 + s} \end{aligned}$$

At $\omega_C = 5$, slope further changed by -20 , there is simple pole.

$$\begin{aligned} \text{Factor} &= \frac{1}{1 + T_3 s}, \text{ where } T_3 = \frac{1}{\omega_C} = \frac{1}{5} = 0.2 \\ &= \frac{1}{(1 + 0.2s)} \end{aligned}$$

Hence the transfer function is

$$G(s)H(s) = \frac{79.8 s^2}{(1 + 2s)(1 + s)(1 + 0.2s)}$$